

Parking entrance staffing study

CSS142P Modelling and Simulation | Group 8 | 30 September 2026

Executive summary

At estimated normal demand, average total delay was 2.57 seconds for A, 1.80 for B, and 1.70 for C. B saved 0.77 seconds per car for 2 additional scheduled guard-hours. C saved 0.87 seconds for 3 additional guard-hours. Extending B to C saved 0.10 seconds; its paired 95% interval for C minus B was [-0.13, -0.08] seconds.

Across 80%, 100%, and 120% demand, the largest mean delay was 3.21 seconds. No spillover was observed in the nine study combinations. This does not prove zero risk. Under these estimates, retain A as the starting staffing choice: the experiment has not established a need for more guards. Additional staffing has a measured time benefit, but its financial value was not modelled. This recommendation applies to the assumed arrangement and inputs. It is not a verified staffing recommendation for the real campus.

This report is a study under explicit assumptions. It does not establish that the revised scope was accepted by the instructor.

Problem, boundary, and inputs

The decision is which of three guard schedules to use under the assumed entrance arrangement. A has 12 scheduled guard-hours, B has 14, and C has 15.

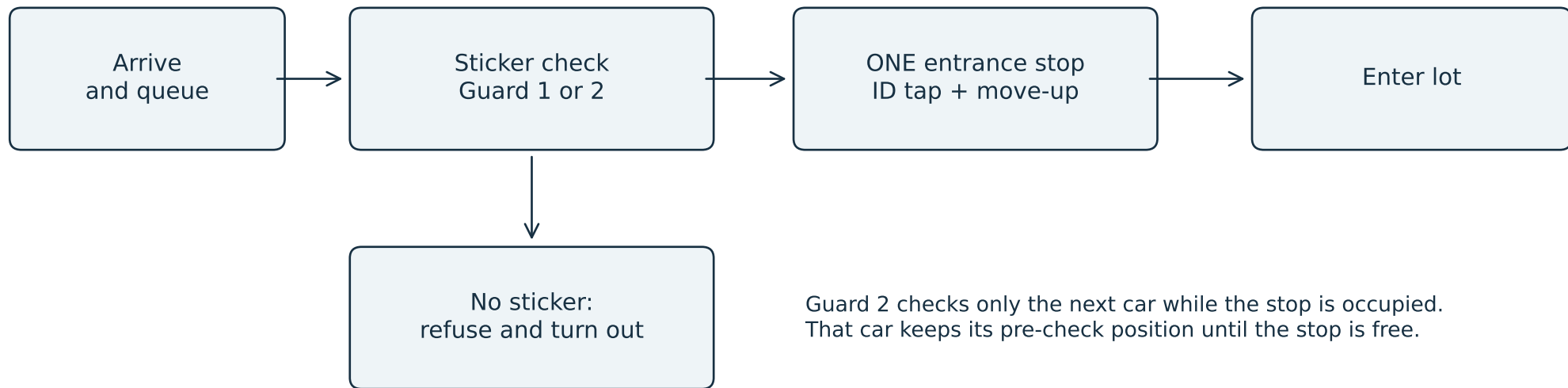
Each operating day starts empty at 07:00. Arrivals stop at 19:00 and remaining cars are processed. The model excludes parking-space limits, exit traffic, abandonment, and reader failures.

Inputs are documented group estimates: approximately 402 arrivals/day, a 100/hour peak, activity means of 4 seconds checking, 4 seconds tapping and barrier operation, 3 seconds moving, and 45 seconds refusal. Activity standard deviations are 2, 2, 1, and 15 seconds. The no-sticker share is 5%; K is 8.

Lognormal activity parameters come from those means and standard deviations. Arrivals use a piecewise-constant Poisson process. Neither assumption is a fit to campus observations.

The revised model replaces independent booths with one stop and a pre-check position. The supplied progress report is the basis for this choice; the physical arrangement remains unconfirmed.

Assumed entrance, not a confirmed campus layout



Method, experiment, and checks

SimPy implements a discrete-event model. A car is admitted after checking, tapping, and moving, or refused after checking and refusal activity. Guard 2 checks only the next car while the stop is occupied; it starts no new check after its scheduled closing.

The experiment crosses three schedules with demand factors 0.8, 1.0, and 1.2. Seeds 1 to 30 give independent days. Each schedule receives identical cars within a seed and demand level. Processing distributions and other inputs remain fixed during demand sensitivity.

Total delay is time in the system minus the car's own activity time. Initial waiting is reported separately. The approach queue includes the pre-check position. Guard utilization counts active checks and refusals, with available time extended to finish final guard activity.

27 recorded checks passed. Hand traces, deterministic and empty cases, conservation, FIFO, queue-area accounting, closing rules, reproducibility, per-car Lindley comparisons, known-answer queues, and reader holding-time bounds are exported in `verification_checks.csv`.

Means, standard deviations, and 95% t intervals describe the simulation output. Paired staffing differences also include Bonferroni intervals across three comparisons per measure and demand. Replication precision is recorded for all nine combinations.

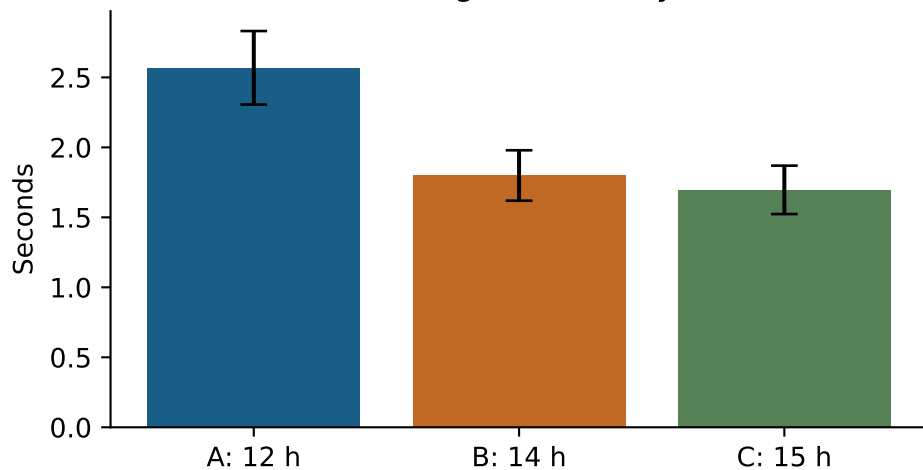
Results at normal demand

Normal demand. 30 simulated days per schedule. Full confidence intervals are in the notebook and CSV exports.

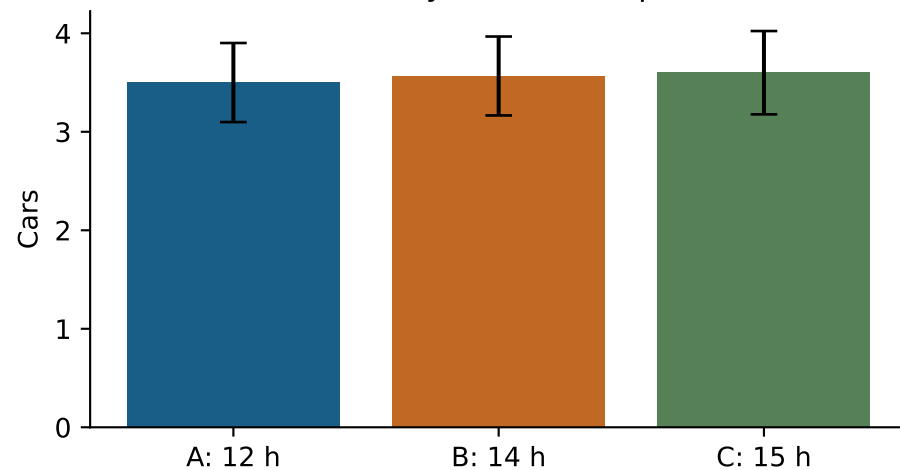
Schedule	Guard-hours	Delay (s)	Max queue	Guard use	Spillover (min)
A	12	2.57	3.50	5.79%	0.00
B	14	1.80	3.57	4.96%	0.00
C	15	1.70	3.60	4.63%	0.00

Estimated normal demand | 30 days per schedule | bars show mean and 95% CI

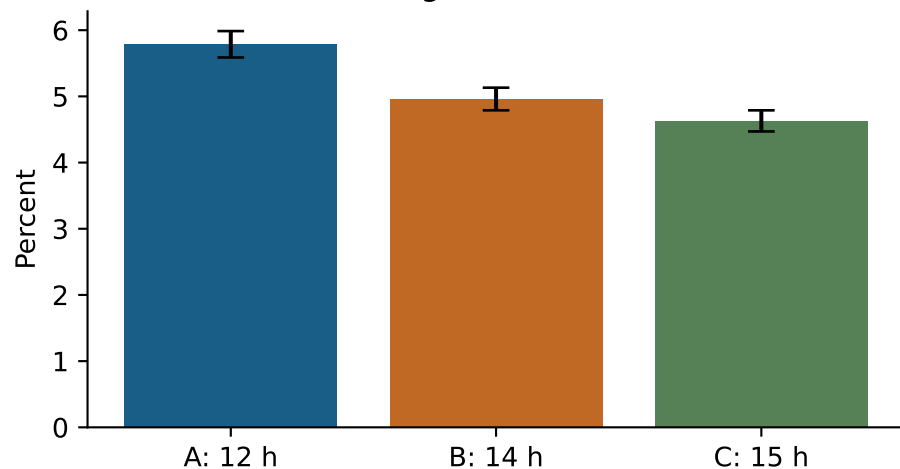
Average total delay



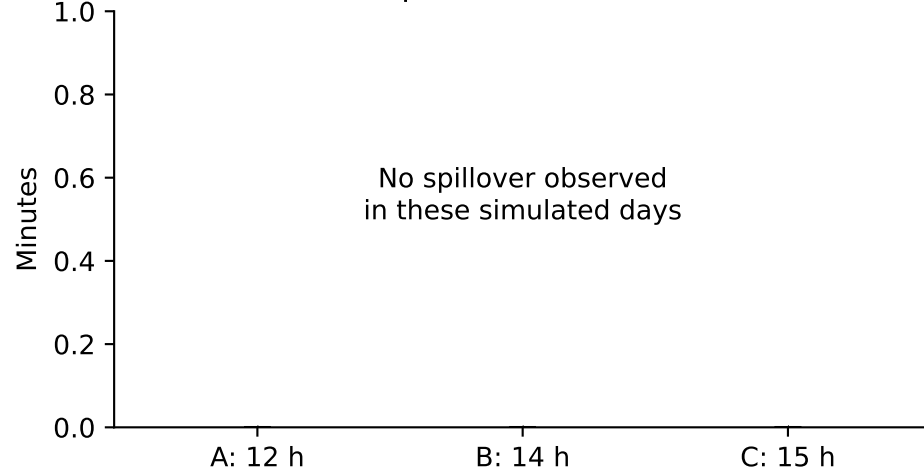
Mean daily maximum queue



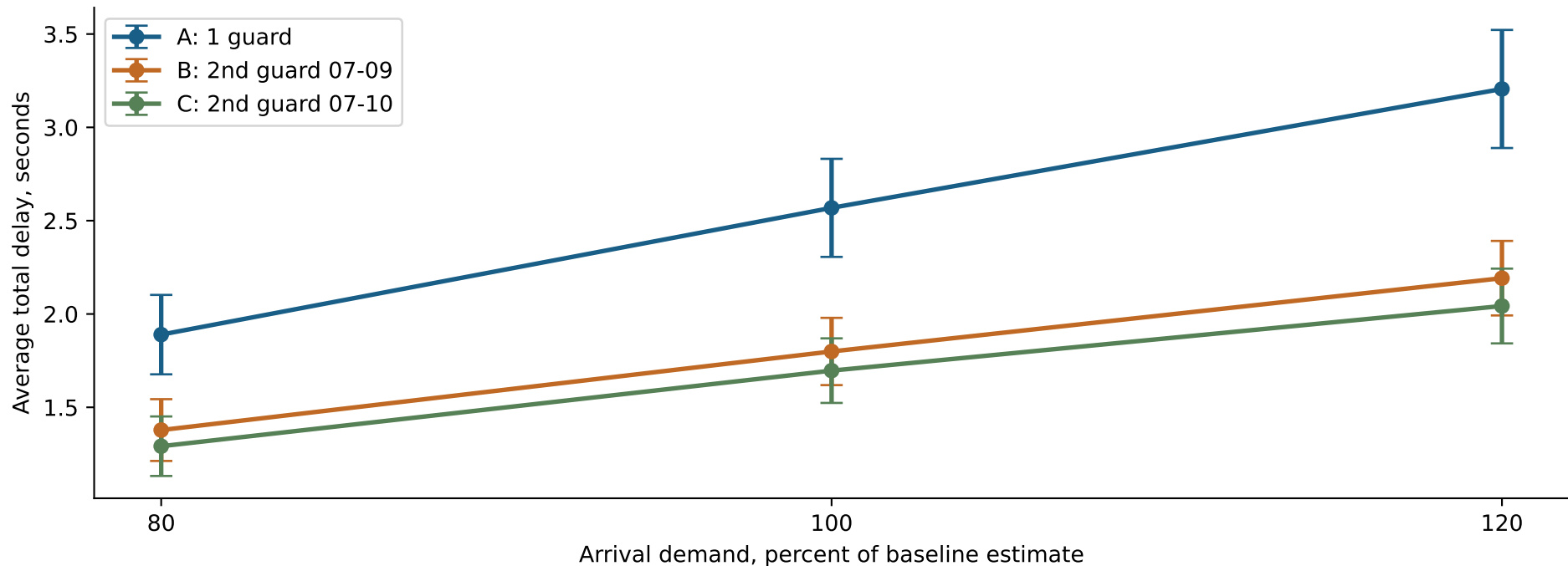
Active guard utilization



Spillover duration



Do the staffing comparisons change when arrivals change?



Validation, recommendation, and limits

Across 80%, 100%, and 120% demand, the largest mean delay was 3.21 seconds. No spillover was observed in the nine study combinations. This does not prove zero risk. Under these estimates, retain A as the starting staffing choice: the experiment has not established a need for more guards. Additional staffing has a measured time benefit, but its financial value was not modelled. This recommendation applies to the assumed arrangement and inputs. It is not a verified staffing recommendation for the real campus.

Field validation was unavailable. Tests of zero or low demand, demand multiplied by five, and a reader ten times slower checked reasonable responses. These cases are not observations of the campus.

The output intervals cover simulation sampling variation. Estimated activity times, arrival volumes, K, refusal behavior, and the shared-reader policy may be inaccurate. Changing these assumptions can change the conclusion.

Uniform demand scaling preserves the timing of the peak. It does not test a peak that moves later or lasts longer. Financial cost and the acceptable waiting-time target were not specified.

The optional public-data downloader and fluid approximation were not used to validate the presented results. No actual street blockage or complaints have been established.

Sources and reproduction

Primary project records: Modeling_Proposal.pdf;
CSS142P_Final_Project_Proposal_Format.pdf;
CSS142P_Final_Project_Requirements_and_Guidelines.pdf; the user's professor
comment; and progress-report.md describing PR #2.

Input estimates and their attributed sources: data/input_parameters.csv and
data/arrival_rates.csv. The input CSV cites a commercial RFID timing range; it
is background attribution, not independently established campus timing. The
model's 4-second tap estimate also includes the barrier cycle.

SimPy documentation: <https://simpy.readthedocs.io/> . NumPy random streams:
<https://numpy.org/doc/stable/reference/random/parallel.html> . SciPy statistical
functions: <https://docs.scipy.org/doc/scipy/reference/stats.html> .

Environment and notebook execution: <https://docs.astral.sh/uv/guides/projects/>
and <https://nbclient.readthedocs.io/en/latest/client.html> . Exact package
versions are recorded in uv.lock.

From the repository root: `uv sync --locked`. Then `uv run --locked python`
`Group8_Parking_Booth_Simulation/tools/verify_notebook.py`. This executes a fresh
notebook copy, checks the exports, and preserves evidence. Add `--publish only` to
refresh the working notebook and generated exports.